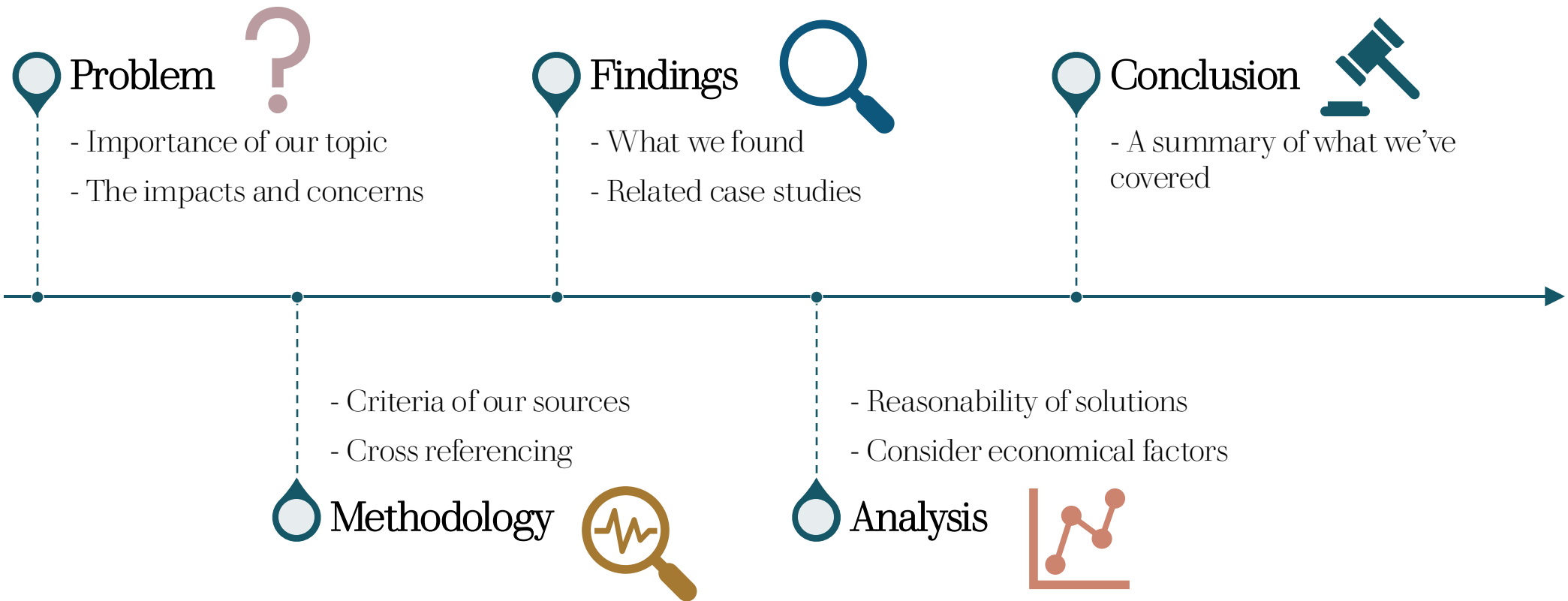


# *RESPONSIBLE & SUSTAINABLE INNOVATION*

BY MARIAM LING SAIF & DAVID  
31<sup>ST</sup> OCTOBER



# INTRODUCTION



# *GLOBAL CRISIS & TECHNOLOGICAL IMPACT*

- Climate crisis is one of our greatest challenges with increasing carbon emissions and environmental degradation
- E-waste crisis is becoming more prominent with rapid tech advancements and shorter product life cycles, increasing year after year
- Energy consumption has been on the rise with the increase of cloud computing and need for large data centres
- Technology plays a major role in all these, however today we'll be focusing on the e-waste crisis





# *THE CONSEQUENCES*

- Environment [5]
  - Toxic Pollution / Harming Ecosystems / Lost of Biodiversity

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- Climate Change
  - Carbon Emissions / Unsustainable

---
- Resource Scarcity
  - Depletion of Resources / Wasted Economic Value

---
- Social Impacts
  - Human Health / Digital Divide

# *METHODOLOGY*



Relevant – has this got to do with our topic at hand



Accurate – how reliable and factful is the information



Alignment – does this data fit in with the rest of our research



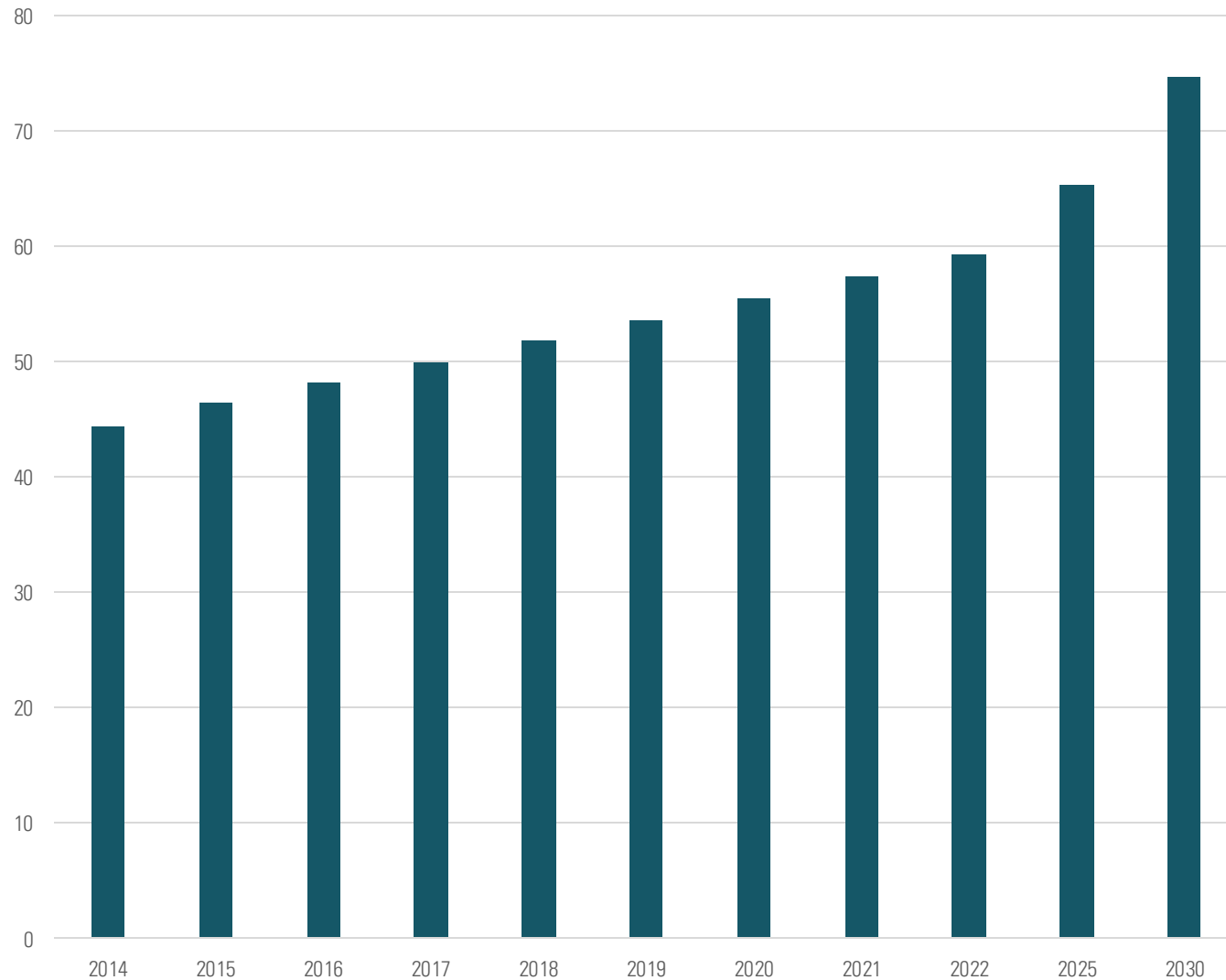
Time/Date – when was it released, is it up to date



Reference – is the data accurate and consistent with other sources



Total Annual Global E-Waste Generation  
(Million Metric Tons) [6]



# *RISE OF E-WASTE*



# *E-WASTE RECYCLING FACTS*

**400,000**

---

Tons of e-waste are  
produced by house holds  
each year

**31.2%**

---

Of E-waste is recycled  
within the UK. What  
about the rest ?

**6 Million**

---

Tons of e-waste  
generated in the UK  
every year

**3600**

---

Homes can be powered  
annually from a million  
recycled laptops

**34kg**

---

Of gold can be recovered  
from recycling a million  
phones

# *CASE STUDY: IMPACT OF FREE AND OPEN-SOURCE SOFTWARE*



- The university uses a free open-source software called Moodle, this reduces paper usage and reduces energy usage.
- Usage of the e learning platform increased from 6% to 7% when the system was first established . The total weight of paper saved amounted to 5854kg. [8]
- The University of Moratuwa has reduced the development effort by 133-man years therefore saving 400GWh in energy. [8]
- Every 5 development years is equivalent to about 1 computer, so it also reduces e-waste by 27 computers. [8]
- The university's open-source software includes a library management system, document management system and other customisations.



# *POTENTIAL SOLUTIONS*

- Green Computing [1]
  - Detoxifying e-waste by promoting proper disposal
  - Increase the life cycle with compatible software
  - Reduce energy usage with efficient code
- Circular Economy [7]
  - Durability & repair by supporting modular upgrades
  - Refurbish products to resell at cheaper prices
  - Closed-loop recycling where materials are re-used
- Smart management [3]
  - Streamline waste sorting with Internet of things
  - Optimise transport by utilising artificially intelligence in decisions
  - Predict roadblocks and optimise operations with machine learning



# *CRITICAL ANALYSIS*



Technology



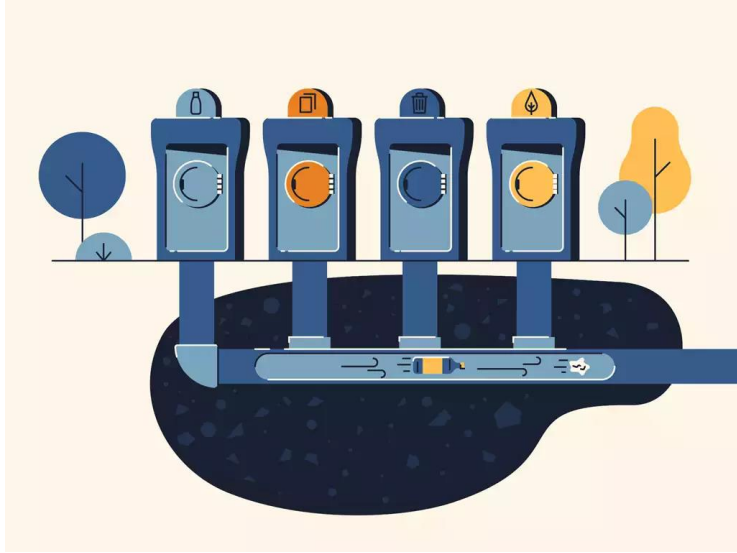
Economics



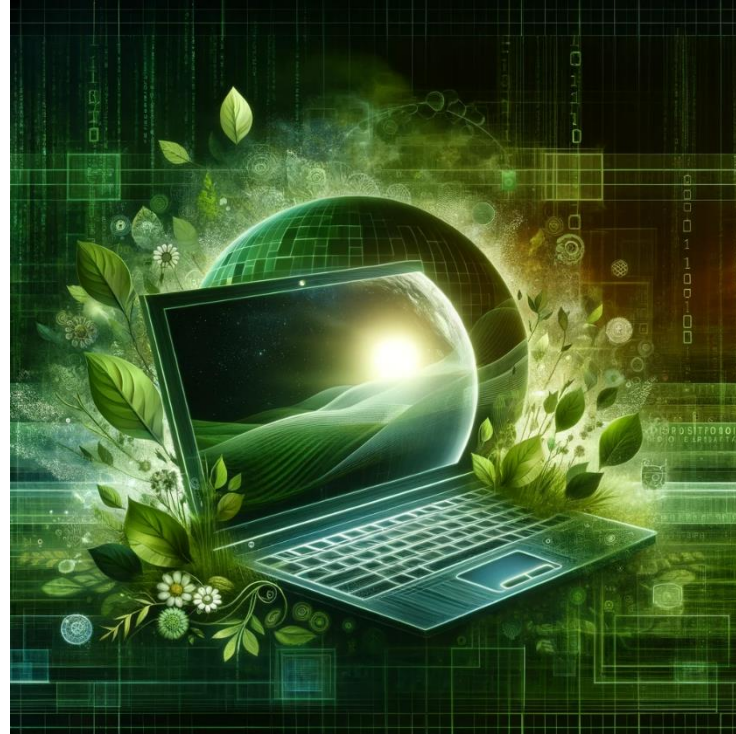
Social



Environment



Smart management



Green computing



Circular Economy

*POTENTIAL SOLUTIONS*

# *RECOMMENDATION*

-> Green Computing <-

- Economically: Reduces the need to upgrade hardware as regularly hence will save both the costs of processing (extraction) and production (manufacturing) from raw materials
- Technologically: We already have the means to achieve green computing with it being in practice already, we just need more involvement from everyone
- Socially: Reduces environmental harm by reducing e-waste and improving the health standard of communities near e-waste disposal sites
- Business: The commitment to more efficient hardware results in reduced costs which will allow businesses to reduce the costs of their services making them more desirable than competitors



# *SOFTWARE ENGINEERING IN REUSE AND RECYCLE*

The Key Takeaways



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- 
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*THANK YOU FOR LISTENING*